

A PROJECT TITLE

ON

Waste Classification for Recycling Using CNN and Transfer Learning

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

Keerthana G 2520090132

Sutte Alekhya 2520090126

Under the Esteemed guidance of

Dr. Y. Lakshmi Prasanna

Assistant professor, CS-IT

Department of CS~~IT~~ & AI-DS



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CS-IT & AI-DS

Name of Title: Waste Classification for Recycling Using CNN and Transfer Learning

1. ABSTRACT:

Waste management has become a major environmental challenge due to the increasing amount of municipal solid waste generated every day. Manual waste segregation is time-consuming, expensive, and often inaccurate, resulting in reduced recycling efficiency. This project proposes an intelligent waste classification system using Convolutional Neural Networks (CNN) and Transfer Learning to automatically identify different categories of waste from images. The system classifies waste into categories such as plastic, paper, cardboard, glass, metal, organic waste, and trash, enabling efficient recycling and reducing environmental pollution.

The proposed model utilizes a pre-trained deep learning architecture such as MobileNetV2, ResNet50, or EfficientNetB0, which has already learned useful image features from large datasets. Transfer learning significantly reduces training time while improving classification accuracy even with limited training data. The dataset consists of labelled waste images that undergo preprocessing techniques such as resizing, normalization, and data augmentation to improve model performance and reduce overfitting.

The trained CNN model extracts visual features automatically and predicts the correct waste category for a given input image. Performance is evaluated using metrics such as accuracy, precision, recall, F1-score, and confusion matrix. Experimental results show that transfer learning achieves higher accuracy than a CNN built from scratch while requiring fewer computational resources.

This project demonstrates how artificial intelligence can support sustainable waste management by automating waste sorting, reducing human effort, and increasing recycling efficiency. The proposed system can be integrated into smart recycling bins, industrial waste sorting plants, and municipal waste management systems. In the future, the model can be enhanced by deploying it on mobile devices or IoT-enabled smart bins for real-time waste classification and environmental monitoring.

2. INTRODUCTION:

Rapid urbanization and population growth have led to a significant increase in waste generation worldwide. Improper disposal of waste causes environmental pollution, health hazards, and inefficient recycling processes. Traditional waste segregation methods rely on manual labour, making them slow, costly, and prone to human error.

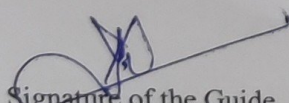
Machine Learning and Deep Learning have revolutionized image recognition tasks by enabling computers to identify objects with high accuracy. Convolutional Neural Networks (CNNs) are among the most effective deep learning models for image classification because they automatically learn important visual features from images.

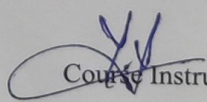
This project uses Transfer Learning, where a pre-trained CNN model such as MobileNetV2 or ResNet50 is fine-tuned on a waste image dataset. Instead of training a deep neural network from scratch, transfer learning reuses knowledge learned from millions of images, resulting in faster training, better performance, and reduced computational requirements.

The proposed system classifies waste into recyclable and non-recyclable categories, helping improve recycling efficiency and supporting sustainable environmental practices. The model can be used in smart cities, recycling industries, educational institutions, and automated waste collection systems.

3. SOFTWARE REQUIREMENTS:

Component	Requirement
Operating System	Windows 10/11 or Linux
Programming Language	Python 3.10 or above
IDE	Google Collab, Jupyter Notebook, VS Code
Deep Learning Framework	TensorFlow, Keras
Libraries	NumPy, Pandas, Matplotlib, OpenCV, Scikit-learn
Dataset	Garbage Classification Dataset / Trash Net Dataset
Hardware	Intel i5/i7 Processor, 8 GB RAM (Minimum), GPU Recommended
Version Control	Git & GitHub (Optional)


Signature of the Guide


Course Instructor